The Æther-flow Interpretation of Relativity

Higher-Derivative Quartic Branch Strict-Branch Decay Test and First Nonlinear-Stability Obstruction

Abstract

The explicit minimal quartic-compatible completion $S_{\min,4}$ now has a first local small-data theorem, a corrected coercive energy, a $T \sim \varepsilon^{-1}$ bootstrap on the same strict auxiliary elliptic-control branch, and a same-branch test showing that $\kappa_\theta > 0$ and $\kappa_\Omega > 0$ are structural for the present local solvability architecture. The next honest question is therefore narrower again: can that corrected strict-branch energy be paired with decay or other integrable nonlinear control strongly enough to yield a genuine small-data nonlinear stability theorem on the same branch?

This note performs that test at the same disciplined scope. It assumes the best linear-compatible decay one would naturally ask for on the current branch: wave-type t^{-1} decay for the Einstein variables and Schrödinger-type $t^{-3/2}$ decay for the quartic scalar channel generated by

$$\partial_t^2 \sigma + \frac{\lambda_4}{m_S^2 M_*^2} \Delta^2 \sigma = 0.$$

Under that decay package, the genuinely propagating quadratic Einstein–quartic-scalar interactions are time-integrable. In particular, the quartic scalar stress

$$\Theta_{ij}^{(\sigma)} = \mathcal{O}\left(\frac{\lambda_4}{M_*^2} D_i D_j \sigma \, \Delta_\gamma \sigma \right) + \mathcal{O}_{\text{l.o.t.}}$$

and the lower-order curvature corrections in the scalar equation do not create the first obstruction.

The obstruction appears earlier and is more rigid. Writing $n \equiv N - 1$, the maximal-gauge lapse equation takes the elliptic form

$$-\Delta_\gamma n + b_N \, n = \mathcal{S}_N,$$

where $\mathcal{S}_N$ contains the nonnegative leading pieces

$$\frac{c_\pi}{m_S^2} \pi_\sigma^2 + c_\Delta \frac{\lambda_4}{M_*^2} (\Delta_\gamma \sigma)^2 + c_K |K|^2.$$

For decaying data with nonzero monopole

$$\mathfrak{m}_N(t) \equiv \int_{\mathbb{R}^3} \mathcal{S}_N(t, y) \, dy,$$

the elliptic inverse produces the standard Coulomb tail

$$n(t, x) = \frac{\mathfrak{m}_N(t)}{4\pi |x|} + \mathcal{O}(|x|^{-2}).$$

The scalar equation still contains the background-slope forcing

$$(\partial_t - \mathcal{L}_\beta)\sigma = N\frac{\pi_\sigma}{m_S^2} + \sigma_0 n,$$

so on the wave zone $|x| \sim t$ one obtains a nonintegrable t^{-1} source unless a new renormalization or monopole cancellation is supplied. Therefore the present branch does *not* yet yield a straightforward upgrade of the corrected coercive bootstrap to a genuine nonlinear stability theorem for decay back to the exact flat background. The first obstruction is long-range auxiliary/gauge feedback through the elliptic lapse, not the quartic principal operator itself.

1. Introduction

The higher-derivative quartic branch has now reached a much sharper nonlinear status than before. The active manuscript line already contains one explicit minimal quartic-compatible nonlinear completion, one reduced Einstein–quartic-scalar $3+1$ system, one first local small-data well-posedness theorem on a strict auxiliary elliptic-control branch, one corrected coercive energy yielding a $T \sim \varepsilon^{-1}$ bootstrap on that same branch, and one same-branch relaxation test showing that the ordered-flow coefficients $\kappa_\theta > 0$ and $\kappa_\Omega > 0$ are structurally tied to the present local solvability architecture.

That changes the meaning of the next step again. The missing question is no longer whether a branch exists, whether a local theorem exists, or whether the strict elliptic assumptions can be weakened by a small variation of the same proof. The next burden is to test whether the corrected strict-branch energy can be combined with decay or other integrable nonlinear control strongly enough to close a genuine small-data nonlinear stability theorem on the branch already recorded.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that boundary. It does not claim a nonlinear stability theorem. It does not claim global instability of the current branch. It performs the next narrower test only: if one supplements the corrected strict-branch energy by the natural linear-compatible decay package, does the bootstrap close? The answer at present scope is no, and the first obstruction can already be isolated.

2. The Strict-Branch Upgrade Test

The strict branch already fixes the reduced variables, gauge, and coercive energy architecture. The only additional issue for a genuine nonlinear stability theorem is whether the nonlinear remainder can be made time-integrable.

Definition 1 (Strict-branch corrected energy)**.** *Let $\mathfrak{F}_N(t)$ denote the corrected coercive functional from the strict-branch bootstrap note, equivalent there to the local control energy $\mathfrak{E}_N(t)$ for $N \geq 6$:*

$$\frac{1}{2}\mathfrak{E}_N(t) \leq \mathfrak{F}_N(t) \leq \frac{3}{2}\mathfrak{E}_N(t). \tag{1}$$

Definition 2 (Linear-compatible decay package)**.** *Fix a strict-branch solution on $[0,T]$. The solution is said to satisfy the* linear-compatible decay package *if there exists $C_* > 0$ such that for every $t \in [0,T]$,*

$$\mathfrak{D}_g(t) \equiv \|\partial(\gamma-\delta)(t)\|_{W^{2,\infty}} + \|K(t)\|_{W^{1,\infty}} + \|\beta(t)\|_{W^{2,\infty}} \leq C_*\varepsilon(1+t)^{-1}, \tag{2}$$

$$\mathfrak{D}_\sigma(t) \equiv \|\pi_\sigma(t)\|_{W^{1,\infty}} + \|D^2\sigma(t)\|_{W^{1,\infty}} \leq C_*\varepsilon(1+t)^{-3/2}, \tag{3}$$

$$\mathfrak{D}_{\mathrm{aux}}(t) \equiv \|Q(t)\|_{W^{2,\infty}} + \|\chi(t)\|_{W^{2,\infty}} + \|W^T(t)\|_{W^{2,\infty}} \leq C_*\varepsilon(1+t)^{-1-\delta_*} \tag{4}$$

for some $\delta_* > 0$.

Remark 1. *Definition 2 is deliberately a* test *package, not a theorem. The wave-type t^{-1} rate in* (2) *is the natural first decay one would seek for the Einstein sector. The scalar rate in* (3) *is the corresponding Schrödinger-type decay suggested by the linear quartic scalar reduction*

$$\partial_t^2\sigma + \frac{\lambda_4}{m_S^2 M_*^2}\Delta^2\sigma = 0,$$

whose dispersion relation is $\omega(\xi) \simeq |\xi|^2$. The point of the present note is not to prove (2)–(4)*; it is to see whether such decay would even be enough.*

3. Propagating Quadratic Terms Are Integrable Under the Test Package

The explicit-completion and local-theorem notes already isolate the leading nonlinear structures that can enter the corrected energy estimate:

1. the quartic scalar stress in the Einstein equations,
2. the curvature-dependent lower-order corrections in the scalar equation,
3. transport and commutator terms involving β, K, and the auxiliary variables.

Under Definition 2 these terms are not the first obstruction.

Proposition 1 (Integrable propagating remainders)**.** *Assume the strict branch, the corrected coercive equivalence* (1)*, and the linear-compatible decay package* (2)–(4)*. Then every quadratic remainder in the commuted strict-branch energy estimate arising from:*

1. *the quartic scalar stress*

$$\Theta_{ij}^{(\sigma)} = \mathcal{O}\Big(\frac{\lambda_4}{M_*^2} D_i D_j\sigma\, \Delta_\gamma\sigma\Big) + \mathcal{O}_{\mathrm{l.o.t.}},$$

2. *the scalar momentum density*

$$J_i^{(\sigma)} = -\pi_\sigma D_i\sigma + \mathcal{O}_{\mathrm{l.o.t.}},$$

3. the scalar lower-order correction

$$\mathcal{N}_\sigma = \mathcal{O}(NK\pi_\sigma) + \mathcal{O}\Big(\frac{\lambda_4}{M_*^2}\mathrm{Ric}[\gamma]\cdot D^2\sigma\Big) + \mathcal{O}\Big(\frac{\lambda_4}{M_*^2}D\,\mathrm{Ric}[\gamma]\cdot D\sigma\Big) + \mathcal{O}_{\geq 2},$$

4. and the already-controlled auxiliary insertions

is bounded by

$$|\mathcal{R}_{\mathrm{prop}}(t)| \leq C_N\Big(\mathfrak{D}_g(t)^2 + \mathfrak{D}_g(t)\mathfrak{D}_\sigma(t) + \mathfrak{D}_\sigma(t)^2 + \mathfrak{D}_{\mathrm{aux}}(t)\big(\mathfrak{D}_g(t) + \mathfrak{D}_\sigma(t)\big)\Big)\mathfrak{F}_N(t). \tag{5}$$

In particular, the coefficient multiplying $\mathfrak{F}_N(t)$ in (5) lies in $L^1_t([0,T])$.

Proof. Each propagating remainder already recorded in the explicit-completion and local-theorem notes vanishes on the flat exact-closure background and contains at least one perturbative coefficient factor together with one differentiated unknown. Under (2)–(4), the metric and auxiliary factors decay at least like $(1+t)^{-1}$, while the scalar principal factors π_σ and $D^2\sigma$ decay like $(1+t)^{-3/2}$. Therefore every product in the displayed list decays at worst like $(1+t)^{-2}$ and often faster. Commuting high-order derivatives through those terms and pairing against the corrected energy variables gives (5) by the same Sobolev multiplication used in the coercive-bootstrap note. □

Corollary 1 (The quartic principal operator is not the first obstruction)**.** *Under Definition 2, the explicit quartic scalar channel and the genuinely propagating Einstein–quartic-scalar couplings do not by themselves prevent an L^1_t-based upgrade of the strict-branch coercive bootstrap.*

Proof. Equation (5) shows that those terms contribute only an integrable coefficient in the corrected energy inequality. □

4. Elliptic Lapse Feedback on the Strict Branch

The obstruction comes from a different place. The scalar equation still contains the background-slope coupling

$$(\partial_t - \mathcal{L}_\beta)\sigma = N\frac{\pi_\sigma}{m_S^2} + \sigma_0(N-1). \tag{6}$$

The issue is therefore the decay of the lapse perturbation $n \equiv N-1$, not the quartic principal operator.

Proposition 2 (Strict-branch lapse equation for the perturbation)**.** *On the strict auxiliary elliptic-control branch, the lapse perturbation $n = N-1$ satisfies an elliptic equation of the schematic form*

$$-\Delta_\gamma n + b_N[\gamma, K, \sigma, \pi_\sigma, \psi, \pi_\psi]\, n = \mathcal{S}_N[\gamma, K, \sigma, \pi_\sigma, Q, \chi, W^T, \psi, \pi_\psi], \tag{7}$$

where $b_N \geq 0$ in the flat-background small-data regime and the source has the expansion

$$\mathcal{S}_N = c_K|K|^2 + \frac{c_\pi}{m_S^2}\pi_\sigma^2 + c_\Delta\frac{\lambda_4}{M_*^2}(\Delta_\gamma\sigma)^2 + \mathcal{S}_N^{\mathrm{div}} + \mathcal{S}_N^{\mathrm{int}}, \tag{8}$$

with constants $c_K, c_\pi, c_\Delta > 0$, a divergence-form remainder $\mathcal{S}_N^{\mathrm{div}} = D_i\mathcal{J}_N^i$, and an integrable remainder $\mathcal{S}_N^{\mathrm{int}}$ that is at least cubic in the perturbation size.

Proof. The local well-posedness note already records that the lapse is determined slice by slice by the maximal-gauge elliptic equation. Subtracting the flat exact-closure background $N = 1$ rewrites that equation in the perturbative form (7). In maximal slicing, the principal source terms are the Einstein quadratic geometry density together with the matter and scalar energy density. The explicit-completion note identifies the scalar energy density pieces

$$\rho_\sigma = \frac{1}{2m_S^2}\pi_\sigma^2 + \frac{\lambda_4}{2M_*^2}(\Delta_\gamma \sigma)^2 + \mathcal{O}_{\text{l.o.t.}},$$

and those contributions enter the lapse source with positive sign in the standard maximal-gauge equation. The remaining lower-order terms either have divergence form or contain at least one extra perturbative factor. Their exact coefficients are unnecessary for the present argument. □

Definition 3 (Lapse monopole)**.** *For a strict-branch slice define the* lapse monopole

$$\mathfrak{m}_N(t) \equiv \int_{\mathbb{R}^3} \mathcal{S}_N(t, y)\, dy. \tag{9}$$

Remark 2. *The current strict-branch identities do not force $\mathfrak{m}_N(t)$ to vanish. Because $\mathcal{S}_N$ contains the nonnegative leading pieces shown in* (8)*, generic small but nontrivial branch data have $\mathfrak{m}_N(t) \sim \varepsilon^2$.*

Theorem 1 (Generic Coulomb tail of the strict-branch lapse)**.** *Assume the strict branch and let $n = N - 1$ solve* (7) *with decaying boundary condition at spatial infinity. If $\mathfrak{m}_N(t) \neq 0$, then for each fixed t,*

$$n(t, x) = \frac{\mathfrak{m}_N(t)}{4\pi|x|} + \mathcal{O}(|x|^{-2}) \qquad \text{as } |x| \to \infty. \tag{10}$$

Consequently, on the wave zone $|x| \simeq 1 + t$,

$$|n(t, x)| \gtrsim |\mathfrak{m}_N(t)|(1 + t)^{-1} \tag{11}$$

unless a nongeneric cancellation forces the monopole to vanish.

Proof. In the strict branch the spatial metric stays uniformly elliptic and H^{N-1}-close to the Euclidean metric. Therefore the Green kernel of the operator $-\Delta_\gamma + b_N$ has the same leading $1/(4\pi|x|)$ asymptotics as the Euclidean Laplacian, with only lower-order corrections. Decompose $\mathcal{S}_N$ into its integral $\mathfrak{m}_N(t)$, the divergence term, and the higher-order remainder. The divergence term contributes only $\mathcal{O}(|x|^{-2})$ to the far field, and the same is true for $\mathcal{S}_N^{\text{int}}$ because it is more localized and has one extra small factor. The monopole part therefore controls the leading tail, giving (10). Evaluating at $|x| \simeq 1 + t$ yields (11). □

5. First Obstruction to the Stability Upgrade

The previous section identifies the only part of the current strict branch that does not automatically improve under the linear-compatible decay test. That is enough to block a straightforward nonlinear stability theorem aimed at decay back to the exact flat background.

Theorem 2 (Failure of the straightforward integrable upgrade test)**.** *Assume a strict-branch solution satisfies the corrected coercive bootstrap and the linear-compatible decay package of Definition 2. Assume also that the lapse monopole is generically nonzero, so that* (11) *holds with $|\mathfrak{m}_N(t)| \gtrsim c_0\varepsilon^2$*

on a nontrivial time interval. Then the scalar forcing term $\sigma_0(N-1)$ in (6) is not time-integrable on the wave zone:

$$\int_0^T \sup_{|x|\simeq 1+t} |\sigma_0(N-1)(t,x)|\, dt \gtrsim |\sigma_0|\, c_0\varepsilon^2 \int_0^T (1+t)^{-1}\, dt. \tag{12}$$

Therefore the current strict-branch package does not *close a straightforward L^1_t-based upgrade of the $T \sim \varepsilon^{-1}$ coercive bootstrap to a genuine nonlinear stability theorem for decay back to the exact flat background.*

Proof. Proposition 1 and Corollary 1 show that every genuinely propagating quadratic remainder is L^1_t under Definition 2. The only remaining nonintegrable candidate in the scalar evolution is the background-slope forcing $\sigma_0(N-1)$. By Theorem 1, the lapse perturbation on the wave zone has size at least $c_0\varepsilon^2(1+t)^{-1}$ whenever the monopole does not vanish. Multiplying by $|\sigma_0|$ and integrating in time gives (12). Hence the current branch does not furnish the integrable forcing needed for a direct global decay bootstrap around the exact flat background in the unrenormalized variable σ. □

Corollary 2 (First obstruction on the present branch)**.** *At current disciplined scope, the first obstruction to promoting the strict-branch corrected coercive bootstrap to a genuine nonlinear stability theorem is long-range auxiliary/gauge feedback through the elliptic lapse equation. More precisely, the obstruction is the Coulombic tail of $N-1$ generated by the nonzero lapse monopole and inserted into the scalar equation through the background-slope coupling $\sigma_0(N-1)$.*

Proof. This is exactly the content of Theorem 2 together with the integrable remainder statement of Proposition 1. □

Remark 3. *Corollary 2 does* not *say that the current branch is globally unstable in principle. It says only that the* straightforward *strict-branch upgrade test fails before one reaches a full theorem. A successful future route would need at least one new ingredient: for example a renormalized scalar unknown that removes the long-range lapse phase, a gauge in which the lapse obeys a different asymptotic equation, or a different completion or auxiliary architecture that changes the source of the elliptic tail itself.*

6. What This Step Establishes

The present manuscript establishes the following.

1. It tests the strict-branch corrected coercive energy against the natural linear-compatible decay package one would first try for a genuine nonlinear stability theorem.

2. It shows that the quartic scalar principal operator and the genuinely propagating quadratic Einstein–quartic-scalar couplings are not the first obstruction under that test.

3. It isolates the first obstruction as the long-range Coulomb tail of the elliptically determined lapse perturbation.

4. It identifies the specific mechanism by which that tail re-enters the evolution: the background-slope forcing term $\sigma_0(N-1)$ in the scalar equation.

It does not establish the following.

1. It does not prove that the present branch can never satisfy any nonlinear stability theorem after further renormalization or reformulation.

2. It does not choose a new completion, a new gauge, or a new auxiliary architecture.

3. It does not reopen the branch-selection problem.

4. It does not alter the earlier exact-closure, local-theorem, coercive-bootstrap, or strict-elliptic-relaxation results already on record.

7. Conclusion

The strict-branch coercive bootstrap has now been tested against the first natural stability-upgrade mechanism rather than merely named as a future task. Under the linear-compatible decay rates one would normally seek on this branch, the propagating quadratic Einstein–quartic-scalar interactions are already integrable. The first obstruction comes instead from the slice-wise elliptic gauge sector: the lapse perturbation acquires a generic Coulomb tail, and the scalar equation feeds that tail directly back through the background-slope term $\sigma_0(N-1)$.

That conclusion sharpens the live program materially. The next burden is no longer to ask vaguely whether "more decay" might help. The first obstruction is now identified. Only after that identification does it become mathematically disciplined to consider a renormalized scalar variable, a different gauge, or a different completion or auxiliary architecture if one wants a genuine theorem beyond the current strict branch.